Volcano ECDLP census, Phase C: analysis and report

Study `research/volcano_ecdlp_census_20261017/`; plan `PLAN.md`, analysis plan `ANALYSIS_PLAN.md` (frozen after data collection and before inspection), amendments `AMENDMENTS.md` (A1, A2). Every number below is read from `analysis/results.json` (written by `analysis/analyze.py`, 10,000 bootstrap or permutation replicates, seeds 20261022 and 20261023). **Status: observed, not established.** No speedup is claimed. Independent validation (`VALIDATION.json`) has not happened yet; this report is the analysis, not its validation.

1. Verdicts

1. **Positive control first (gates everything): PASS.** On the V3 (j = 0) crater, mean S(`rho-cubic`) / mean S(`rho-neg`) = **0.5733**, 95% CI [0.5619, 0.5850] over 4000 paired runs (theory 1/sqrt(3) = 0.5774; pass needs the CI upper bound below 0.85). The detector sees a 42% cost reduction, so the 'equivalent' verdicts below are informative, not relabelled 'uninformative'.
2. **Level effect (V1, V2; 5 arms; 35 level ratios): 33 equivalent, 0 different, 2 inconclusive.** TOST margins are +-5% for rho and BSGS and +-10% for IC. The two inconclusive ratios are both `rho-plain` at level 1 (CI not inside the margin; neither differs from 1 after Holm). No method shows a level effect on either volcano. With one volcano per l, this is an observed absence of an effect at these margins, not an established one (PLAN §5: replication on a second volcano of the same l is missing).
3. **Curve effect against the isomorphic negative control:** no variance-ratio test survives Holm (smallest adjusted p = 0.130); BSGS is degenerate (identical S for a curve and its re-scalings). 1 curve is flagged in total (V2, `rho-plain`, vertex 69), and it is not flagged for the other arms.
4. **Transfer:** every one of the 403168 measured runs recovered k (95% lower bound 0.9963 for the smallest cell). BSGS counts are identical across D, Ua, Ub and S in 100% of triples. Paired source ratios for rho-neg, rho-plain and ic: 23 of 24 equivalent, 1 inconclusive, 0 different. Transport is cheap against solving (section 6) and no 'transport, then solve' ratio is a speedup signal (0 signals).
5. **IC against matched `rho-neg` (distance remaining, not pass/fail):** IC costs 42.4x (V1) and 61.5x (V2) the rho cost in S with nothing priced, 65.1x and 92.7x with every uncharged operation priced at 1 GAE on both arms. The order (rho ahead) holds across the whole 0 / 0.1 / 1 band at every level of V1, V2 and V3. No level improves IC's position.
6. **Wall time: not admitted for any (V, arm).** The A/A ratio is within [0.95, 1.05] on all three volcanoes, but only 87% to 94% of rows reached isolation L2, so condition 1 fails everywhere. Wall tables are descriptive.

2. What was measured

- 79 audited ecbench sessions (V3, V1, V2; V4 is censused, not measured, A1.1), each `audit OK` with exact replays; 403,168 measured runs after dropping 265,952 warm-up runs. Unverified measured runs: 0.
- Arms: `rho-neg` (matched reference), `rho-plain`, `bsgs`, `bsgs-neg`, `ic` (`ic-fb64-mitm`, frozen in `FROZEN.json`), `rho-neg-aa` (A/A), `rho-cubic` (V3 crater). Families D, Ua, Ub, S, N are isogenous; `ctl` (two isomorphic re-scalings per vertex) is the negative control and never pooled.
- Population: V1 (l = 2, depth 4, 400 curves per family), V2 (l = 3, depth 3, 378), V3 (j = 0, depth 4, 16; descriptive). Curves per level V1 25/25/50/100/200, V2 14/28/84/252, V3 1/1/2/4/8.
- Unit: S (ecbench normalised group operations). Host: shared 4-vCPU cloud VM; binary `ac4bc5b2...` from a clean commit (`runs/progress.jsonl` header); campaign wall about 12.2 h of sessions plus a 5-day pause.
- Statistics: cluster bootstrap over **vertices**, then rows, stratified by level (A2.3). Rows per curve are many; levels are compared on their curves, not their solves.

3. Positive control

V3 crater (census vertex 0), D, N and ctl rows pooled (3 generators; `rho-cubic` was run on one of the two re-scalings), 4000 paired runs. Ratio 0.5733, 95% CI [0.5619, 0.5850] (pair bootstrap); slug-cluster sensitivity [0.5588, 0.5851]. Result: **pass**.

4. Level effect: R(L) = mean S at level L / mean S at the crater

Cells: ratio [95% CI], verdict (E equivalent, D different, I inconclusive). Margins [0.95, 1.053] for rho and BSGS, [0.90, 1.111] for IC. Holm family (a) has 35 tests.

V1 (l = 2)

arm	L1/L0	L2/L0	L3/L0	L4/L0	CV of S per row	achieved 95% half-width of R
rho-neg	0.992 [0.955, 1.032] E	0.997 [0.956, 1.040] E	1.006 [0.972, 1.043] E	1.002 [0.970, 1.036] E	0.46	0.037
rho-plain	1.016 [0.973, 1.062] I	0.997 [0.955, 1.042] E	0.987 [0.951, 1.026] E	0.994 [0.960, 1.030] E	0.47	0.040
bsgs	1.000 [0.991, 1.010] E	1.000 [0.991, 1.010] E	1.000 [0.992, 1.008] E	1.000 [0.993, 1.007] E	0.15	0.009
bsgs-neg	1.000 [0.986, 1.014] E	1.000 [0.986, 1.014] E	1.000 [0.988, 1.012] E	1.000 [0.989, 1.011] E	0.22	0.013
ic	0.980 [0.949, 1.012] E	0.981 [0.949, 1.015] E	0.984 [0.956, 1.013] E	1.011 [0.985, 1.038] E	0.35	0.030

V2 (l = 3)

arm	L1/L0	L2/L0	L3/L0	CV of S per row	achieved 95% half-width of R
rho-neg	0.993 [0.958, 1.030] E	0.984 [0.952, 1.016] E	0.998 [0.972, 1.025] E	0.46	0.032
rho-plain	1.030 [0.994, 1.069] I	1.007 [0.973, 1.044] E	1.019 [0.989, 1.051] E	0.46	0.035

bsgs	1.000 [0.989, 1.012] E	1.000 [0.990, 1.010] E	1.000 [0.991, 1.009] E	0.20	0.010
bsgs-neg	1.000 [0.983, 1.017] E	1.000 [0.985, 1.015] E	1.000 [0.987, 1.013] E	0.29	0.015
ic	0.990 [0.944, 1.038] E	0.996 [0.958, 1.036] E	0.983 [0.948, 1.019] E	0.45	0.040

BSGS ratios equal 1 exactly: BSGS cost depends on r and k only, and r is the same on every curve of a volcano. Their bootstrap interval reflects only the spread of the scalars k_i. Secondary mixed model (`log S ~ C(level) + C(family) + (1 | curve)`, REML) agrees: level ratios between 0.977 and 1.038; ICC between 0.0000 and 0.0074 (no curve-level variance beyond the solve noise); non-converged fits: V2|bsgs-neg (reported; the bootstrap is the result). Per-family R(L) (descriptive) are in `results.json` under `R_by_family_descriptive`.

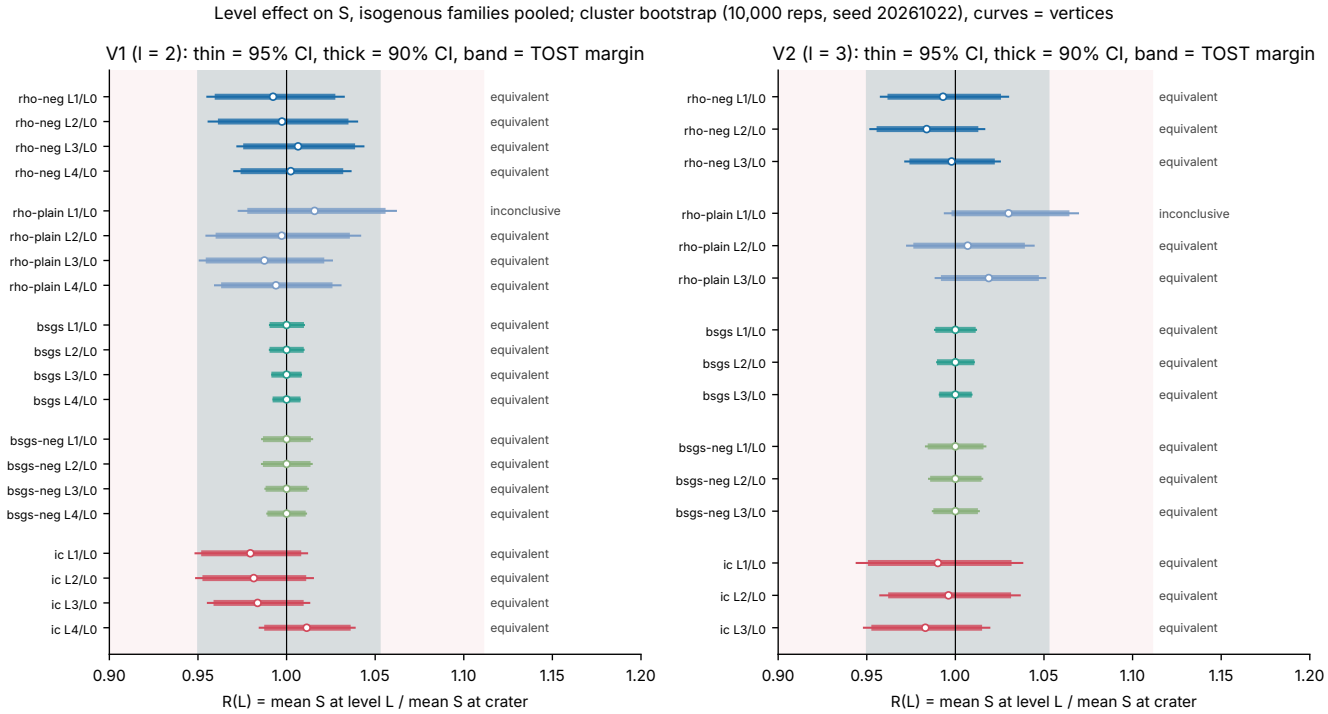


Figure 1. Level ratios with TOST bands. Source: `analysis/results.json` `level_effect`; session ids in `runs/progress.jsonl`.

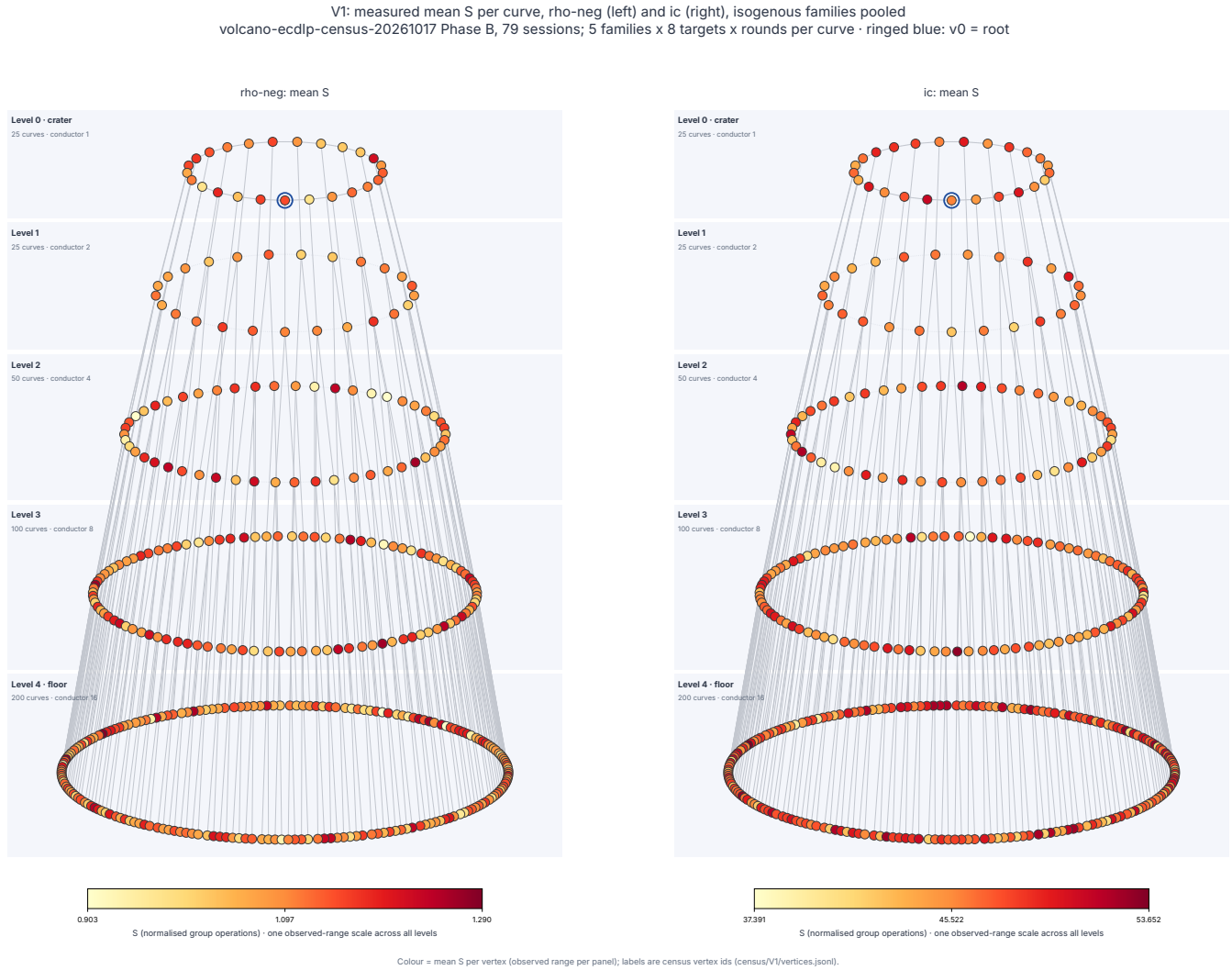


Figure 2a. V1 per-curve mean S for `rho-neg` and `ic` (standard `volcano-diagram/v2`). V2 and V3 are `fig2_volcano_V2/V3`. Colour scales are per panel; the spread within a level is the same size as between levels.

5. Curve effect against the negative control

volcano	arm	vertices	F (between / within)	permutation p	Holm p (across 10)	flagged curves (Holm p < 0.01)
V1	rho-neg	400	0.867	0.926	1.000	none
V1	rho-plain	400	1.032	0.676	1.000	none
V1	bsgs	400	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V1	bsgs-neg	400	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V1	ic	400	0.949	0.800	1.000	none
V2	rho-neg	378	0.900	0.888	1.000	none
V2	rho-plain	378	1.113	0.022	0.130	vertex 69 (level 2, z 4.55, adjusted p 1.0e-04)
V2	bsgs	378	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V2	bsgs-neg	378	degenerate	n/a	n/a	none (D and both re-scalings give identical S)
V2	ic	378	0.945	0.862	1.000	none

F below 1 means the original curves spread no more than their own isomorphic re-scalings. The flagged V2 curve (vertex 69) has z by arm: rho-neg z = −0.24, rho-plain z = 4.55, ic z = 2.91. One curve of the 2,334 tested by the plan's rule (3 non-degenerate arms x 778 vertices), in one arm only, and the corresponding variance-ratio test is not significant after Holm; it is listed, not interpreted.

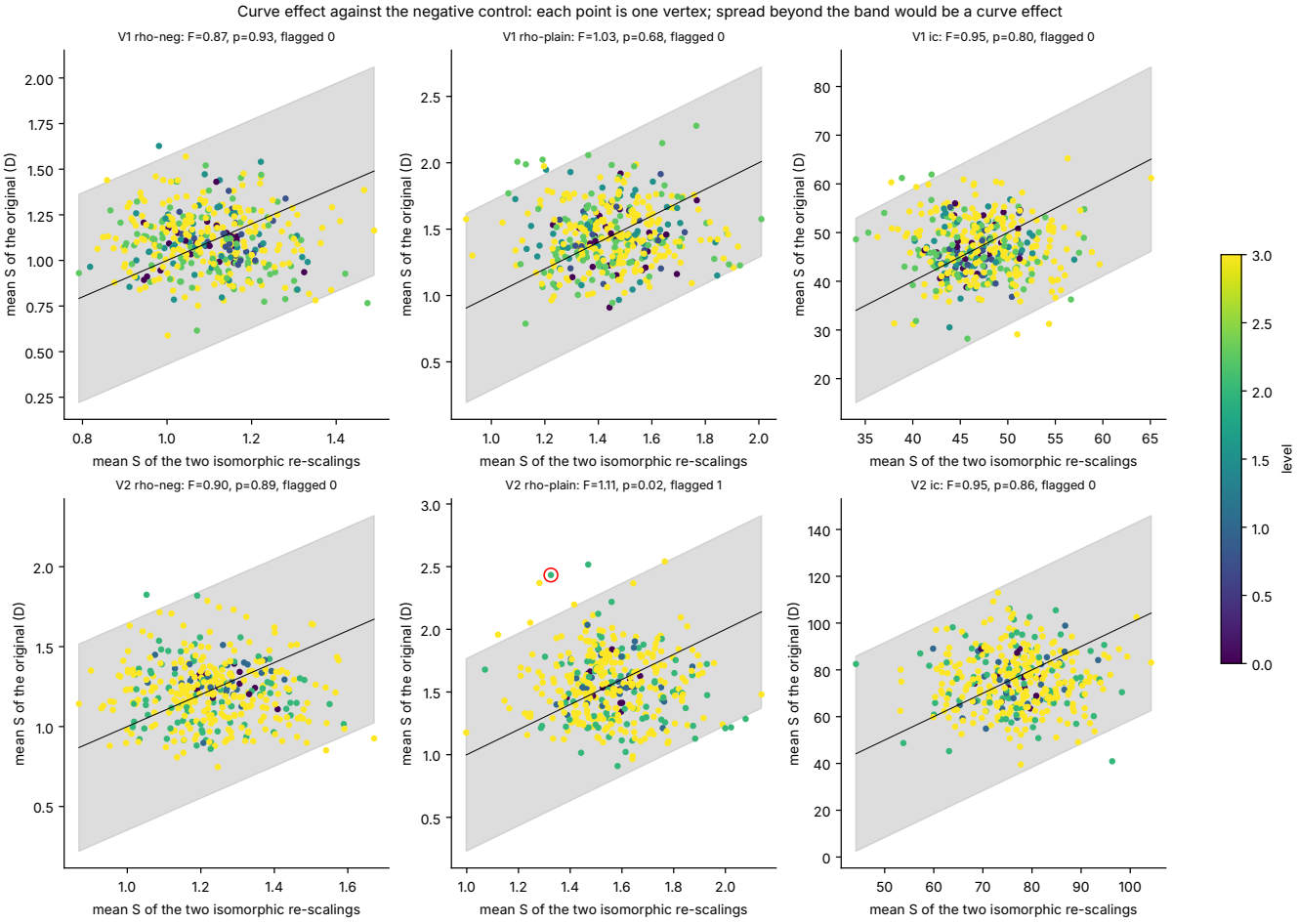


Figure 4. Mean S of each original curve (D) against the mean of its two re-scalings. The grey band is the normal-approximation 99% band of the control, for display only; flags come from the permutation test.

6. Transfer

Recovery. 403,168 of 403,168 measured runs verified (every (V, family, arm) cell 100%; the Clopper-Pearson 95% lower bound is 0.9963 for the 1,000-run cells and higher for larger ones). `matches_transported` is true for all 3,158 transport-benchmark paths.

Paired source ratios (mean S of the source family over D, on shared k_i; 95% CI; TOST margin as in section 4). Nothing is 'different'.

volcano	arm	Ua/D	Ub/D	S/D	N/D
V1	rho-neg	0.996 [0.969, 1.024] E	0.997 [0.969, 1.027] E	0.997 [0.969, 1.026] E	0.989 [0.962, 1.017] E
V1	rho-plain	1.018 [0.989, 1.049] E	0.992 [0.963, 1.022] E	1.028 [0.998, 1.059] I	1.021 [0.990, 1.052] E
V1	ic	1.010 [0.988, 1.033] E	1.002 [0.979, 1.026] E	0.991 [0.968, 1.015] E	0.997 [0.975, 1.019] E
V2	rho-neg	1.007 [0.978, 1.036] E	1.003 [0.976, 1.032] E	1.008 [0.979, 1.037] E	1.014 [0.985, 1.043] E
V2	rho-plain	0.995 [0.965, 1.025] E	1.004 [0.974, 1.034] E	1.003 [0.973, 1.034] E	1.001 [0.972, 1.031] E
V2	ic	0.999 [0.968, 1.030] E	1.015 [0.985, 1.047] E	1.001 [0.971, 1.031] E	1.017 [0.990, 1.046] E

N uses another scalar sequence (A1.3), so its pairing is nominal. BSGS identity check (fraction of (vertex, target, round) triples with equal `ops_gae` across the families present): V1 bsgs 1.000 over 3600; V1 bsgs-neg 1.000 over 3600; V2 bsgs 1.000 over 3584; V2 bsgs-neg 1.000 over 3584; V3 bsgs 1.000 over 2040; V3 bsgs-neg 1.000 over 2040.

Transport cost (`runs/transfer_cost.jsonl`, 200 repetitions per path, `ca volcano --transport-bench`; field-operation counts are a model, not instrumented). Per path length d, median ns per path (IQR) and Velu point maps (2 per edge):

volcano	d = 1	d = 4	d = 8	d = max
V1	2.8 us (2.7 to 2.8), d = 1, 2 maps	11.3 us (11.0 to 12.2), d = 4, 8 maps	22.3 us (22.1 to 22.7), d = 8, 16 maps	60.3 us (56.6 to 62.9), d = 20, 40 maps
V2	3.1 us (3.0 to 3.1), d = 1, 2 maps	12.5 us (12.3 to 12.7), d = 4, 8 maps	25.3 us (24.7 to 27.6), d = 8, 16 maps	44.7 us (40.4 to 45.3), d = 13, 26 maps
V3	3.2 us (3.2 to 3.3), d = 1, 2 maps	12.8 us (12.3 to 13.1), d = 4, 8 maps	n/a	19.7 us (19.0 to 20.3), d = 6, 12 maps

Transport as a fraction of the destination solve (median `solve_wall_ns` of the arm at that level; wall is descriptive, section 8):

volcano	arm	median fraction over levels (max over paths)	Velu maps / mean ops_gae at 0.1 / 1 GAE per map
V1	rho-neg	0.35% (1.03%)	0.055% / 0.55%
V1	bsgs-neg	0.38% (1.12%)	0.058% / 0.58%
V1	ic	0.06% (0.18%)	0.001% / 0.01%
V2	rho-neg	0.27% (1.21%)	0.092% / 0.92%
V2	bsgs-neg	0.29% (1.29%)	0.102% / 1.02%
V2	ic	0.09% (0.39%)	0.001% / 0.01%

Transport, then solve, against solve where born (A2.3; per source family, arm and destination level, priced at 0 / 0.1 / 1 GAE per Velu map): 240 cells; the CI excludes 1 at all three prices in 14 of them, and **0 meet the plan's speedup condition** (excludes 1 on the lower side at all three prices and disjoint from the native-instance control). The cells that exclude 1 are V1 `ic` from S (ratio about 1.2, above 1), V3 `ic` from D and Ua (about 1.2, above 1; the crater is cheaper for IC, section 9) and V2 `rho-plain` from Ua (about 0.6, **below 1**). The V2 `rho-plain` Ua cell is the same at every level including level 0 (no transport at all), and the paired Ua/D ratio for it is equivalent to 1, so it reflects a high mean at the single source vertex (one floor curve, few rows), not a transport benefit; its CI overlaps the native-instance control CI (about 0.75 to 1.34), so it is not a signal under the plan's condition. The price band moves any ratio by at most 1.5%. Transport is at most about 1.3% of a solve in wall time for any path (medians below 0.5%) and at most about 1.3% in GAE at 1 GAE per Velu map (V2, `bsgs-neg`), far less for IC.

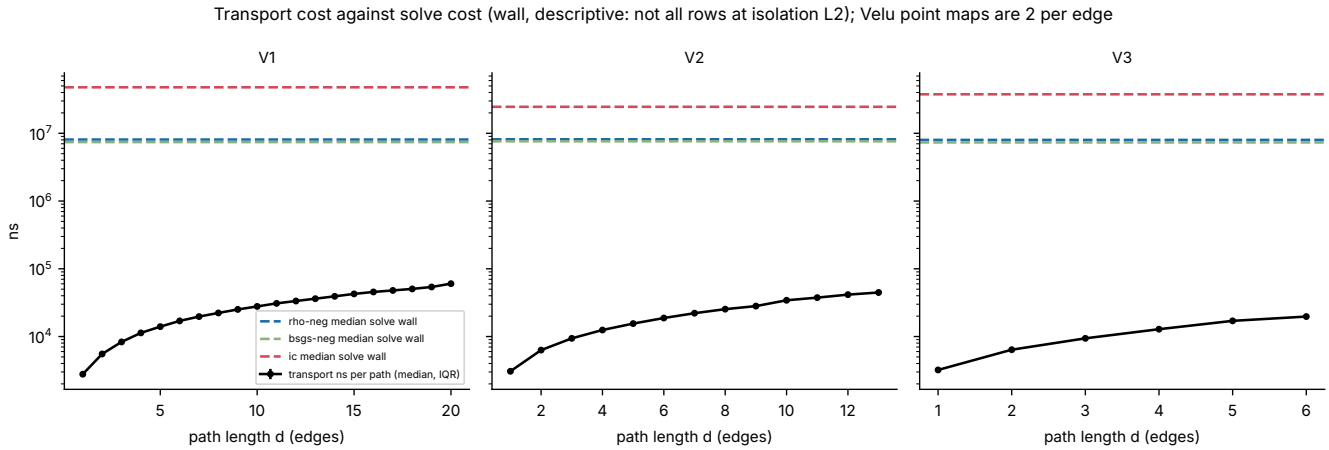


Figure 5. Transport ns per path against median solve wall by path length (descriptive wall).

7. IC against matched rho (distance remaining)

Mean S(ic) / mean S(rho-neg), isogenous families pooled, 95% CI, with every uncharged native operation (`ic`: lookups, target-guard probes, row operations, Legendre symbols, square roots; `rho-neg`: canonicalisations) priced at 0 / 0.1 / 1 GAE on **both** arms. Counts come from the run records (A2.1). Both arms still omit native work ecbench does not name, so both are lower bounds.

V1

level	curves	price 0	price 0.1	price 1	order across the band
0	25	42.6 [41.0, 44.4]	46.6 [44.8, 48.5]	65.4 [62.8, 68.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	25	42.1 [40.8, 43.4]	46.0 [44.6, 47.4]	64.6 [62.5, 66.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	50	41.9 [40.3, 43.6]	45.8 [44.1, 47.6]	64.4 [61.8, 67.0]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
3	100	41.6 [40.6, 42.6]	45.5 [44.4, 46.6]	63.9 [62.3, 65.5]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
4	200	43.0 [42.2, 43.8]	47.0 [46.2, 47.8]	66.0 [64.8, 67.2]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
all		42.4 [41.9, 42.9]	46.4 [45.8, 46.9]	65.1 [64.3, 65.9]	

V2

level	curves	price 0	price 0.1	price 1	order across the band
0	14	61.9 [59.4, 64.5]	67.3 [64.6, 70.2]	93.3 [89.3, 97.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	28	61.7 [59.1, 64.5]	67.1 [64.3, 70.2]	93.0 [88.9, 97.4]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	84	62.7 [60.7, 64.7]	68.2 [66.0, 70.4]	94.5 [91.4, 97.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
3	252	61.0 [59.9, 62.2]	66.3 [65.1, 67.6]	91.9 [90.1, 93.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)

all		61.5 [60.6, 62.4]	66.9 [65.9, 67.9]	92.7 [91.3, 94.1]	

V3 (descriptive: one curve at levels 0 and 1, 2 / 4 / 8 above)

level	curves	price 0	price 0.1	price 1	order across the band
0	1	49.0 [47.7, 50.3]	53.3 [51.9, 54.7]	73.3 [71.3, 75.3]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
1	1	59.9 [58.3, 61.5]	65.2 [63.4, 67.0]	90.1 [87.6, 92.7]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
2	2	62.6 [59.4, 66.0]	68.2 [64.6, 71.9]	94.4 [89.3, 99.6]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
3	4	62.0 [60.0, 64.0]	67.5 [65.3, 69.6]	93.3 [90.3, 96.4]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
4	8	60.5 [58.6, 62.4]	65.8 [63.8, 68.0]	91.0 [88.1, 94.0]	rho-neg ahead of ic at all three prices (95% CI above 1 at each)
all		58.8 [57.8, 59.8]	64.0 [62.9, 65.1]	88.4 [86.9, 89.9]	

Uncharged operations per run (V1): `ic` 292,972 (about 1.9 per charged GAE), `rho-neg` 3,239 (about 0.90 per charged GAE). The gap to rho is therefore about 43x unpriced and 65x at 1 GAE (V1), a distance of more than an order of magnitude at this size and no level of any volcano changes it. This is the measured baseline gap, not a result for or against IC's potential.

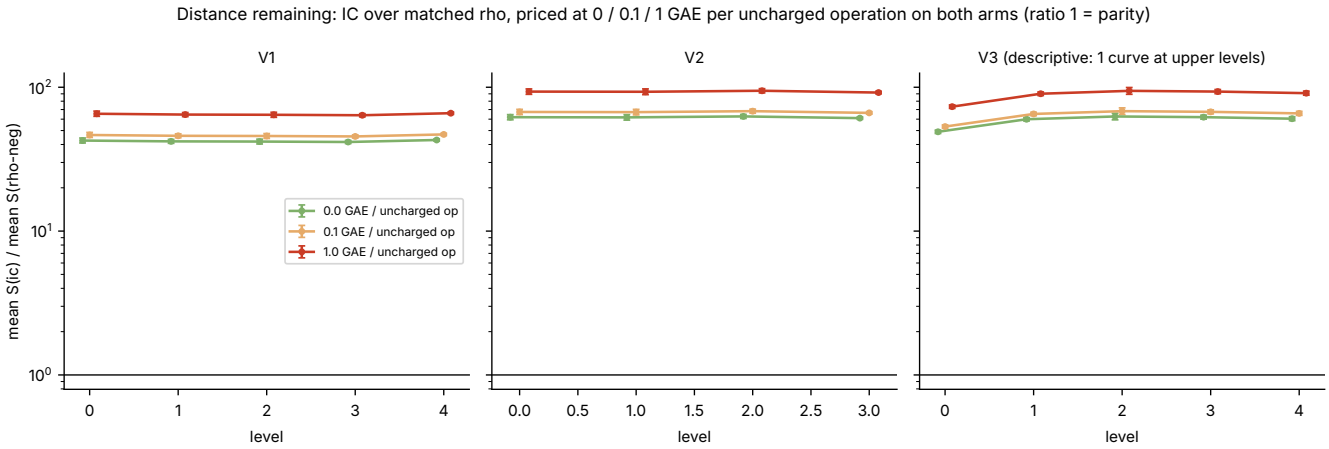


Figure 3. IC / rho-neg by level, three prices, 95% CI (log scale).

8. Wall time

volcano	A/A ratio (rho-neg-aa / rho-neg) [95% CI]	within [0.95, 1.05]	rows at L2 (rho-neg / ic)	admitted
V1	0.9994 [0.9944, 1.0044]	yes	92.0% / 86.8%	no (isolation L1 rows present)
V2	1.0005 [0.9959, 1.0051]	yes	93.4% / 91.6%	no (isolation L1 rows present)
V3	0.9978 [0.9924, 1.0032]	yes	92.4% / 89.4%	no (isolation L1 rows present)

So wall numbers (`solve_wall_ns` ratios in `results.json`, `R.<level>.wall`) are **descriptive (not admitted)**. For orientation they also show no level trend. The A/A result says the instrument's own noise floor is within 1% pooled.

9. V3 (j = 0), descriptive

One curve at levels 0 and 1, so no between-curve inference. Mean S relative to the crater:

arm	L1	L2	L3	L4
rho-neg	1.003	0.969	0.992	1.003
rho-plain	0.991	0.980	0.997	0.994
bsgs	1.014	1.014	1.014	1.014
bsgs-neg	1.022	1.022	1.022	1.022
ic	1.226	1.238	1.254	1.237

Observed, not tested: `ic` is about 22 to 25% more expensive at every non-crater V3 level than at the j = 0 crater, while the rho and BSGS arms show nothing. This fits IC using the 6 automorphisms of j = 0 on the crater, as the negation-folded oracle does for 2; it is one curve, one volcano and descriptive here, and it makes IC's gap to rho smaller on that curve (49x against about 60x at 0 GAE), not close to parity.

10. CV and achieved precision

CV and achieved precision (mean over levels); BSGS is deterministic in r and k, so its CV is the scalar's spread only

volcano	arm	curves per level	rows	CV of S per row	95% half-width of level mean	95% half-width of R(L)	ICC (mixed model)	converged
V1	rho-neg	25/25/50/100/200	18000	0.46	2.3%	0.037	0.000	yes
V1	rho-plain	25/25/50/100/200	18000	0.47	2.5%	0.040	0.000	yes
V1	bsgs	25/25/50/100/200	18000	0.15	0.6%	0.009	0.000	yes
V1	bsgs-neg	25/25/50/100/200	18000	0.22	0.8%	0.013	0.000	yes
V1	ic	25/25/50/100/200	18000	0.35	1.9%	0.030	0.004	yes
V2	rho-neg	14/28/84/252	17920	0.46	2.2%	0.032	0.002	yes
V2	rho-plain	14/28/84/252	17920	0.46	2.2%	0.035	0.000	yes
V2	bsgs	14/28/84/252	17920	0.20	0.7%	0.010	0.000	yes
V2	bsgs-neg	14/28/84/252	17920	0.29	1.0%	0.015	0.000	no
V2	ic	14/28/84/252	17920	0.45	2.5%	0.040	0.007	yes

Figure 6. Per (volcano, arm): curves per level, rows, per-row CV of S, achieved 95% half-width of the level mean and of R(L), and ICC. Rho CV about 0.46 (calibration 0.40, plan 0.52), IC 0.35 to 0.45, BSGS 0.15 to 0.29. The achieved 95% half-width of R(L) is 3 to 4% for rho and IC and about 1% for BSGS, so the +-5% rho margin and the +-10% IC margin were within reach, as the `rho-plain` level-1 results show (they sit just outside).

11. Limits

- Shared 4-vCPU cloud VM; wall time is descriptive, 8 to 13% of rows were at isolation L1.
- One volcano per l (V1 l = 2, V2 l = 3): no replication on a second volcano of the same l, so nothing can be ‘established’, only observed.
- V3 is descriptive (one curve per upper level, A1.2); its positive control uses one of the two ctl re-scalings for `rho-cubic`.
- V4 (twist of V1) is censused, not measured (A1.1).
- 5-day pause in Phase B (driver died after V2-S-L3-c0, resumed 2026-10-22); the gap is excluded from wall; no result depends on it.
- A1 amendments (V4, V3 shape, target seeds, rho CV, Drive, storage, clean binary) and A2 amendments (this phase, post hoc) shape the analysis; each interpretation is listed in `AMENDMENTS.md`. `ANALYSIS_PLAN.md` was frozen after data collection but before anyone inspected a Phase B cost summary. A 100-replicate smoke run of the code was inspected before the final run. `results.json` records the SHA-256 of `AMENDMENTS.md` as it stood at the final run, before A2.3 and A2.4 were appended.
- Curves are census vertices; the five isogenous families share a vertex and are pooled. A different clustering would change widths, not point estimates.
- IC and rho both leave unnamed native work unpriced; the 0 / 0.1 / 1 GAE band covers only counters ecbench names. Field-operation counts for Velu maps are a model.
- Small sizes only (26-bit group orders); no statement about cryptographic sizes, no speedup, no `claim`.
- The analysis is Python (the plan names it); AGENTS.md prefers native code for results. The validator should re-derive the headline numbers independently.

12. Files

- Code: `analysis/analyze.py`, `analysis/extract_uncharged.py`, `analysis/figures.py`, `analysis/build_report.py`; output `analysis/results.json` (all statistics, n, seeds, SHA-256 of every input), `analysis/figures/*.svg|png`, `analysis/analyze.log`.
- Inputs: `runs/extract/*.csv.xz`, `runs/uncharged/*.csv.xz`, `runs/slugmap/*.json`, `census/V*/`, `runs/transfer_cost.jsonl`, `runs/progress.jsonl`, sessions archives `sessions/*.tar.xz`.
- Copies for browsing: `benchmarks/20261017-volcano-ecdlp-census/` (no `benchmarks/README.md` row until the validation gate is passed).